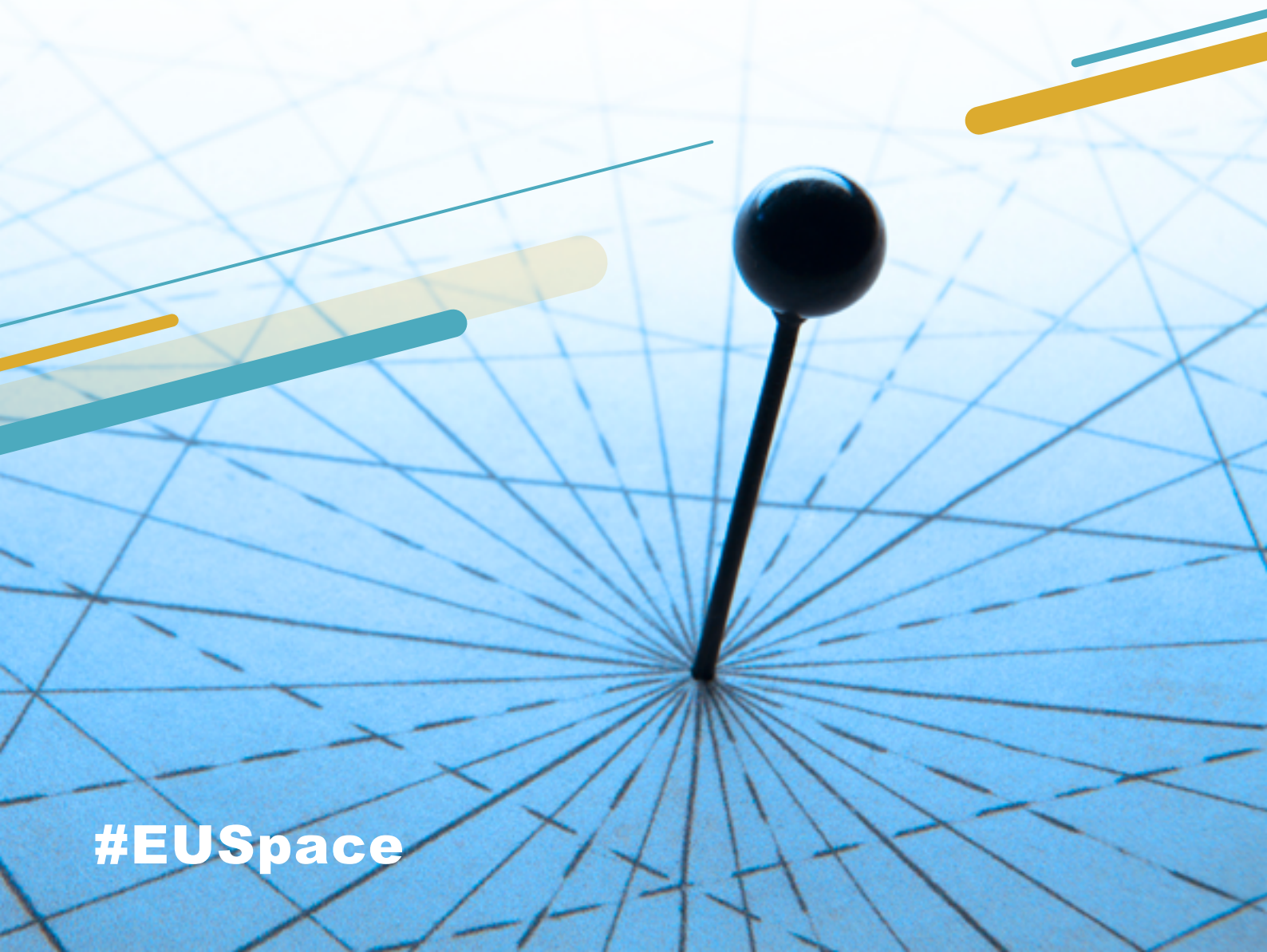
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GALILEO HIGH ACCURACY SERVICE REFERENCE USER ALGORITHM (**HAS UA**) SOFTWARE - USER MANUAL

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1 INTRODUCTION

The Galileo High Accuracy Service (HAS) is an advanced service provided by the Galileo satellite navigation system, aimed at delivering highly precise positioning data to users globally. Unlike standard satellite navigation services, which offer meter-level accuracy, the HAS is designed to provide positioning accuracy at the decimetre level. This is achieved through the transmission of precise correction data, which users can apply to their positioning calculations to significantly enhance accuracy [1]. HAS is particularly beneficial for applications requiring high precision, such as precision agriculture, engineering surveying, autonomous vehicles, and various scientific research activities. By offering free access to high-accuracy positioning, the Galileo HAS enables users to perform tasks that demand a higher level of exactness without the need for additional ground-based augmentation systems.

The service operates by broadcasting corrections via the Galileo satellites, which users can receive and process through compatible receivers and algorithms, such as the HAS User Algorithm detailed in [2]. The HAS User Algorithm (HAS-UA) emerges as a pivotal tool designed to meet these demands by offering enhanced precision in positioning solutions. HAS-UA has been implemented in MATLAB and this user manual serves as a comprehensive guide, offering detailed insights into the installation, configuration, and effective utilization of the HAS-UA.

The HAS-UA is engineered to cater to the needs of users across various sectors that require precise geolocation data. These sectors include advanced geospatial surveying, autonomous vehicle navigation, agricultural management, and construction, where precision down to a few decimetres can be crucial. By implementing sophisticated mathematical models and leveraging precise correction data, the HAS-UA provides users with the capability to perform both Precise Point Positioning (PPP) and Single Point Positioning (SPP). This dual functionality ensures that the algorithm can be adapted to suit a wide range of accuracy requirements and operational contexts.

The HAS-UA tool has been implemented in MATLAB version 2024b, which is the recommended version for conducting software testing. To run the software, the user must have MATLAB version 2022b or higher installed. The HAS-UA is built on a robust and modular framework that supports data ingestion, algorithmic computation, and result generation. The use of MATLAB not only enhances computational efficiency but also provides an accessible platform for users with varying technical expertise to engage with the algorithm. The core functionality is organized into a structured folder system, allowing for easy navigation and execution of the algorithm's components.

This manual delves into the specifics of the algorithm, offering users step-by-step instructions on configuring the software to meet their unique needs. It includes detailed descriptions of configuration files, usage examples, and output interpretation, empowering users to harness the full potential of the HAS User Algorithm. By understanding these elements, users can optimize their data processing workflows, ensuring the delivery of accurate and reliable positioning data.

1.1 Scope of this document

This document provides a brief user manual for using the HAS-UA MATLAB software v1.0.0.

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2 DOWNLOAD AND INSTALLATION

HAS-UA software can be downloaded from the HAS page of the Galileo Service Centre website.

2.1 HAS-UA MATLAB content

The HAS-UA library has a folder structure as shown in Figure 1, which consists of a main folder containing two subfolders: "conf" and "src".

- The "conf" subfolder holds some example configuration files for running the HAS-UA algorithm. These files will be described in further detail in Section 3.2, allowing users to understand their purpose and usage. The configuration files can be placed in any place of the user folder structure, ascertained that the proper configuration file path is provided to the software at execution; it is not mandatory to place them here.
- The "src" subfolder contains all the MATLAB code for the HAS-UA, encompassing the core functionality and implementation of the algorithm. The "src" folder is a comprehensive repository of the HAS-UA's functional components, containing different subfolders that include a range of essential functions, such as data ingestion, algorithmic computations, and result generation, which are critical to the HAS-UA's operation. The organization of these functions within the subfolders provides a structured and modular framework for the HAS-UA's implementation, allowing for efficient and effective execution of the algorithm. Additionally, the "src" folder includes the primary launch file, named *HASUA_Start.m* and described in Section 3.1, which serves as the central entry point for initiating the processing sequence. This launch file enables users to easily execute the HAS-UA algorithm, facilitating a seamless and efficient execution of the algorithm.

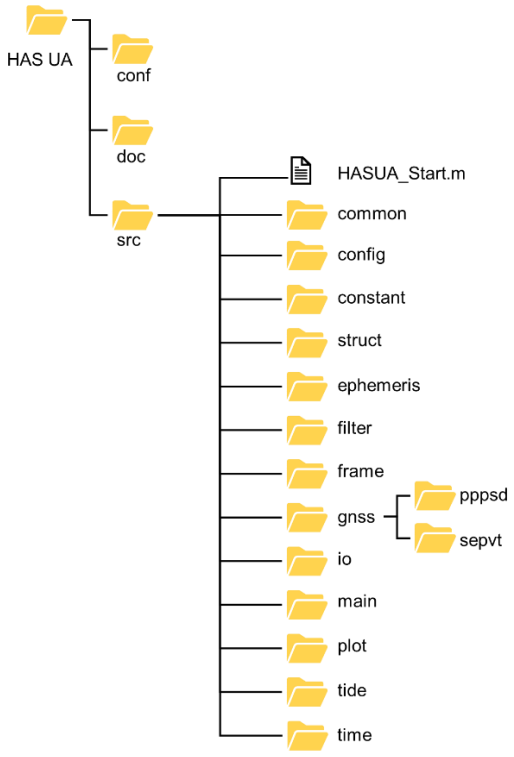


Figure 1: HAS UA folder structure.

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3 HAS-UA USAGE

3.1 Data preparation

As initial step, the user must procure the input data and organise them in a file structure for the processing. For basic HAS processing, the following files are required:

- RINEX observation file v3.04 containing the GNSS measurements, including pseudorange, carrier phase, and Doppler shifts (Doppler shift measurements are optional measurements and their usage can be enabled or disabled as described in 3.2.2);
- RINEX navigation file v3.04 containing the ephemerides broadcast by Galileo and GPS;
- Galileo HAS correction file in BKG NTRIP Client (BNC) RTCM text format (details in Section 3.1.1);
- antenna information file for the satellites and receiver in ANTEX format (antenna information can be retrieved from one or two files, as described in Section 3.2.1).

Before processing, the user must define the folder structure where the data is stored. For example, RINEX observation and navigation (if from the specific receiver) files can be organized in a folder structure as:

- a main folder named as the station (STAT);
- inside the main folder, folders for each year of the data, (YYYY);
- within the year folder, folders for each day named as the corresponding Day of Year (DOY).

The folder structure would result in something like “data\STAT\YYYY\DOY”. Aggregated RINEX navigation (e.g., like IGS BRDC files) and Galileo HAS correction files can be stored in a “common” folder, like “data\common\YYYY\DOY”, as they are not linked to a specific station. The ANTEX files can instead be placed in a structure like “data\common\GPSW”, where “GPSW” stands for the GPS week number, as they are mostly updated on a weekly basis. ANTEX file for GNSS satellites should be retrieved from the HAS Metadata web page in the Galileo Service Centre (GSC) website. This folder structure is merely an example, and users may organize the data as they prefer. It's crucial that the data location is accurately configured in the configuration file, as detailed in Section 3.2.1. The defined folder serves as the input location where necessary files for processing should be placed.

Users can also designate an output folder, which may be the same as the input observation folder or different. The output folder must be specified in the configuration file as described in Section 3.2.1. Within the output folder, users will find a .mat file containing the complete MATLAB processing workspace, a .sol file with a reduced number of fields in text format and a .log file with warnings or errors related to the processing, as detailed in Section 3.2.4.

3.1.1 Differences between SIS and IDD correction file formats

The HAS-UA SW accepts as input an RTCM text file with the SSR HAS corrections. The format of the file is the one described in Section 2.8 “Broadcast Corrections” of the BNC help file. There is a small but fundamental difference between the correction files when the source is the Signal in Space (SIS) or the Internet Data Distribution (IDD) channels of the HAS service. The HAS-UA SW detect the source of the data from the “Mountpoint, source/stream indicator” filed of the file: if any of the strings “E6”, “HASIS” or “HASJRC” is found in the field, the origin is marked as from SIS, otherwise the source is marked as from IDD.

While the SSR RTCM text file is fully compliant with the format described in the BNC help page, when the data are received from SIS and encapsulated in such type of file, the field indicating originally the “SSR message update interval indicator” indicates the “Validity Interval Index” of the SIS message. The index must

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be converted to the validity interval using the “Validity Interval Index Table” provided in the HAS SIS Interface Control Document (ICD) [1] and not the “update interval indicator” conversion provided in the BNC help file (as normally done for files containing data from IDD or other NTRIP sources).

3.2 Configuration File

After procuring the necessary input file, the configuration file must be accurately filled with the data information and processing options. The HAS-UA configuration file is specifically designed to control the data processing parameters. It is structured into four main sections:

- Data Analysis options;
- GNSS options;
- Filtering options;
- Output options.

The configurable elements and the meaning of the fields within each section are detailed below.

3.2.1 Data Analysis options

The complete list of Data Analysis options with their descriptions and comments is provided in Table 1. An example of the Data Analysis option part in a configuration file is shown in Figure 2.

Parameter	Description	Values / Notes
data_dir	The path to the primary directory containing the required data files.	
site_name	The name of the receiver site for which the data is being processed.	
measurements	Path to the RINEX observation file.	If not provided, the system searches in the data_dir.
broadcast	Path to the RINEX navigation file.	If not provided, the system searches in the data_dir.
ssrcorr	Path to the RTCM SSR correction file. This plain ASCII file can be created using the BKG Ntrip Client (BNC) software.	If not provided, the system searches in the data_dir. The format description is available in [3].
antex	Path to the Antenna Exchange Format (ANTEX) file for antenna information.	If not provided, the system searches in the data_dir. User antenna data is always read from this file.
satantex	Path to the ANTEX file specifically for GNSS satellites.	If not specified, satellite data is not automatically detected (in this case, antex must point to a file that includes also satellite data).
outdirname	The name of the folder for output files.	This folder is created inside the data_dir.
outfiledate	Flag to determine if the execution start date and time are appended to output file names.	1: yes (append date/time). 0: no.
start_time	Specifies the first epoch to be processed.	0: reads start time from the RINEX observation file. 1: reads start time manually from the config file (YYYY/MM/DD hh:mm:ss format).

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Parameter	Description	Values / Notes
end_time	Specifies the last epoch to be processed.	Same options and format as start_time.
t_interval	Processing time interval, expressed in seconds.	0: uses the observation interval reported in the RINEX observation file.
t_reset	Time interval (in seconds) for periodic filter resets.	
usematinput	Flag to enable input data from an existing .mat file in data_dir.	0: data taken from the input files. 1: data loaded from the existing .mat file.

Table 1: Data Analysis option list.

```
# Data analysis options ********************************************************************************************************************
data_dir      =      ../data/MET3/2023/358                                          % data directory, the directory should includes all
site_name     =      MET3                                                           % receiver name, (the observation file should includ
measurements  =      ../data/MET3/2023/358/MET300FIN_R_20233580000_01D_30S_MO.rnx % path to the observation RINEX file (if not specifi
broadcast     =      ../data/common/2023/358/BRDM00DLR_S_20233580000_03D_MN.rnx   % path to the navigation RINEX file (if not specifie
ssrcorr       =      ../data/common/2023/358/HASJRC358.23C                        % path to the RTCM SRR correction file (if not speci
antex         =      ../data/common/2023/358/igs20.atx                            % path to the ANTEX file: must contain at least the
satantex      =      ../data/common/2023/358/has14_2345.atx                       % path to the ANTEX file for GNSS satellites (if not
outdirname    =      results_E1_E5a_L1_L2_SIS                                     % name of the folder containing output files (create
outfiledate   =      1                                                            % append (0: no 1: yes) the date and time of the exe

start_time    =      0  2023/12/24 00:30:00                                       % start time (0: from obs, 1: from opt, then YYYY/MM
end_time      =      0  2023/12/24 23:30:00                                       % end time (0: from obs, 1: from opt, then YYYY/MM/D
t_interval    =      0                                                            % processing time interval (0: from obs, NON-ZERO: s
t_reset       =      0                                                            % periodic filter reset time interval (0: no reset,
usematinput   =      0                                                            % flag to enable input of data from mat file found i
```

Figure 2: Example of Data Analysis options in a HAS-UA configuration file.

3.2.2 GNSS options

The complete list of GNSS options with their descriptions and comments is provided in Table 2. An example of the GNSS option part in a configuration file is shown in Figure 3.

Parameter	Description	Values / Notes
solving_mode	Specifies the PVT (Position, Velocity, Time) solving mode.	1: SEPVT. 2: PPPSD. Mode 2 performs also SEPVT first.
sateph	Satellite ephemeris and clock type used.	1: uses only the Broadcast Navigation Message. 2: uses Broadcast message refined with HAS corrections. The HAS corrections are applied as described in [1].
useDoppler	Controls the use of Doppler shift measurements.	0: do not use. 1: use. 2: invert the sign of the measurements.
usePhaseBias	Enables/disables the use of phase bias corrections.	0: disable. 1: enable.

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Parameter	Description	Values / Notes
signalList	Specifies the signals used in processing (Galileo and GPS). (1) S1F1, (2) S1F2, (3) S1F3, (4) S2F1, (5) S2F2, (6) S2F3.	0: Not used 1: L1/E1 2: L2 5: E5a/L5 7: E5b 6: E6 8: E5a+b. The first three fields refer to Galileo (S1) while the remaining three are for GPS (S2). Numbers follow RINEX 3.xx code observation format [3].
sigPriorEm	Priority list for Galileo measurements of signal Em from specific tracking modes to distinguish individual signals. m = 1, 5a, 5b, 5ab, 6.	If not defined, the default list is used. Letters follow RINEX 3.xx code observation format [3].
sigPriorLn	Priority list for GPS measurements of signal Ln from specific tracking modes to distinguish individual signals. n = 1, 2, 5.	If not defined, the default list is used. Letters follow RINEX 3.xx code observation format [3].
elmin	Elevation mask angle (in degrees).	Measurements from satellites with visibility elevation angle lower than this are excluded.
cn0min	Minimum accepted C/N0 (Carrier-to-Noise Density Ratio).	Observations with lower C/N0 values are excluded.
cn0ratioexp	Exponent k of the CN0_min over CN0 ratio $\left(\frac{\mathrm{CN0_{min}}}{\mathrm{CN0}}\right)^k$ of the mapping function combining elevation and CN0.	$k \geq 0$.
maxintfreqpha	Maximum allowed difference for carrier phase measurements on different bands (in meters).	If difference is larger than this, measurements are discarded.
maxintfreqdpl	Maximum allowed difference for Doppler shift measurements on different bands.	If difference is larger than this, measurements are discarded.
maxmeasuregap	Maximum time interval (in seconds) without measurements before the navigation filter is reset.	≥ 0.
mintracking	Minimum time for a satellite being tracked to become pivot (seconds).	≥ 0.
mincssamples	Minimum number of samples after a cycle slip for a satellite to become pivot.	≥ 0.
prevtracked	Satellite required to be tracked at previous sample (1) or not (0) to become pivot.	0: do not require. 1: require. Important if mintracking < measurement sampling rate).

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Parameter	Description	Values / Notes
exclsateclips	Enables/disables the exclusion of satellites currently in eclipse.	0: disable exclusion. 1: enable exclusion.
exclafterecli	Duration (in seconds) a satellite must be excluded *after* exiting eclipse.	Only active if exclsateclips is enabled.
excludeGALsat	List of Galileo satellites to be excluded from processing.	
excludeGPSsat	List of GPS satellites to be excluded from processing.	
tropopt	Troposphere model correction options for SEPVT (1) and PPPSD (2).	0: no correction. 1: Saastamoinen model [4]. 2: SBAS MOPS model [5].
tropmap	Troposphere mapping function for SEPVT (1) and PPPSD (2).	1: Neil mapping function [6]. 2: GCAT tropospheric model [7].
trowest	Zenith wet troposphere parameter estimation for SEPVT (1) and PPPSD (2).	0: do not estimate. 1: estimate.
tidecorr	Options for Earth tide correction in PPP. (1) solid tides; (2) ocean loading; (3) pole tide; (4) solid tides permanent deformation	0: off. 1: on, Solid tides option must be active to perform any of the other corrections
satellitepco	Satellite Phase Center Offset (PCO) correction for SEPVT (1) and PPPSD (2).	0: off. 1: on,
satellitepcv	Satellite Phase Center Variation (PCV) correction for PPPSD.	0: off. 1: on,
receiverpco	Receiver antenna PCO correction for SEPVT (1) and PPPSD (2).	0: off. 1: on,
receiverpcv	Receiver antenna PCV correction for PPPSD.	0: off. 1: on,
phasewindup	Carrier phase wind-up correction for PPPSD.	0: off. 1: on,
shapirodelay	Gravitational delay (Shapiro delay) correction for PPPSD.	0: off. 1: on,
antdelsrc	Source of receiver antenna reference point correction.	0: from observation file. 1: from option file.
antdel	Receiver antenna reference point correction (m) in the ENU frame.	Only used if antdelsrc = 1.
csthresmw, csthresgf, csthresir, csthresdi	Parameters for CSD thresholds using Melbourne-Wubbena, geometry-free, ionosphere rate, and Doppler integration methods, respectively.	Refer to [2] for details.
cs_l_gf, csmax nosn, csminsn, maxcscyc	Additional parameters for general Cycle Slip Detection and Repair.	Refer to [2] for details.

Table 2: GNSS option list.

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```
# GNSS options ************************************************************************************************************************************
solving_mode  =      2                                  % PVT solving mode (1:SEPVT, 2:PPPSD)
sateph        =      2                                  % satellite ephemeris/clock type (1: broadcast ephemeris, 2: HAS ephemeris)
useDoppler    =      1                                  % use Doppler measurements (0: no, 1: yes, 2: invert sign)
usePhaseBias  =      0                                  % use phase bias corrections (0: no, 1: yes)
allSignals    =      0                                  % use in EKF only satellites with all the signals received (0: no, 1: yes)
signalList    =      1,7,0,1,2,0                        % 6 numbers selecting the signal code utilization [1]S1F1 [2]S1F2 [3]S1F3 [4]S2F1 [
                                                        % the digits 1-2-3 indicate the first, second and third signal codes for Galileo (0
                                                        % the digits 4-5-6 indicate the first, second and third signal codes for GPS (xxx00
                                                        % It lists the signal ID (0: Not used, 1: L1/E1, 2: L2, 5: E5a/L5, 7: E5b, 6: E6, 8
                                                        % Code type utilization priorities as for sigPriorXN below
                                                        % Examples:
                                                        % 1,5,0,0,0,0: Galileo E1-E5a
                                                        % 1,7,0,0,0,0: Galileo E1-E5b
                                                        % 1,6,0,0,0,0: Galileo E1-E6
                                                        % 1,5,6,0,0,0: Galileo E1-E5a-E6
                                                        % 1,7,6,0,0,0: Galileo E1-E5b-E6
                                                        % 0,0,0,1,2,0: GPS L1-L2
                                                        % 1,5,0,1,2,0: Galileo E1-E5a, GPS L1-L2
                                                        % 1,7,0,1,2,0: Galileo E1-E5b, GPS L1-L2
                                                        % 1,6,0,1,2,0: Galileo E1-E6, GPS L1-L2
                                                        % 1,5,6,1,2,0: Galileo E1-E5a-E6, GPS L1-L2
                                                        % 1,7,6,1,2,0: Galileo E1-E5b-E6, GPS L1-L2

#sigPriorE1    =      CXB                               % priority list of E1 signals; if not defined here, the default list is used (CXBZ
#sigPriorE6    =      CXB                               % priority list of E6 signals; if not defined here, the default list is used (CXBZ
#sigPriorE5a   =      QXI                               % priority list of E5a signals; if not defined here, the default list is used (QXI
#sigPriorE5b   =      QXI                               % priority list of E5b signals; if not defined here, the default list is used (QXI
#sigPriorE5ab  =      QXI                               % priority list of E5ab signals; if not defined here, the default list is used (QX
#sigPriorL1    =      CPWLXS                            % priority list of L1 signals; if not defined here, the default list is used (CPWL
sigPriorL2    =      LWXSCPD                           % priority list of L2 signals; if not defined here, the default list is used (CPWLX
#sigPriorL5    =      QXI                               % priority list of L5 signals; if not defined here, the default list is used (QXI)

elmin         =      5                                 % GNSS measurement minimum elevation mask angle (deg)
cn0min        =      20                                % lowest accepted GNSS measurement CN0 (dbHz)
cn0ratioexp   =      3                                 % exponent k of the CN0_min over CN0 ratio (CN0_min/CN0)^k of the mapping function
maxintfreqpha =      100                               % largest allowed difference (m) for carrier phase measurements on different bands
maxintfreqdpl =      10                                % largest allowed difference (m/s) for Doppler measurements on different bands
maxmeasuregap =      30                                % maximum time interval (s) without measurements before resetting the NAV filter

mintracking   =      10                                % minimum time for a satellite being tracked to become pivot (seconds), >= 0
mincssamples  =      2                                 % minimum number of samples after a cycle slip for a satellite to become pivot, >=
prevtracked   =      1                                 % satellite required to be tracked at previous sample (1) or not (0) to become pivo

exclsateclips =      1                                 % exclude/keep GNSS satellites in eclipse condition (0: keep, 1: exclude)
exclafterecli =      1800                              % time (s) to exclude the satellite after eclipse (if exclsateclips activated)
excludeGALsat =      0                                 % comma separated list of GAL satellites PRN (e.g.: 1,26,23) to be excluded (0: use
excludeGPSsat =      0                                 % comma separated list of GPS satellites PRN (e.g.: 1,26,23) to be excluded (0: use

tropopt       =      2,2                               % troposphere model correction (0: off, 1: Saastamoinen, 2: MOPS SBAS) [1]SEPVT [2]
tropmap       =      1,1                               % troposphere mapping function (1: Niell, 2: GCAT) [1]SEPVT [2]PPPSD
trowest       =      0,1                               % zenith wet troposphere parameter estimation (0: off, 1: on) [1]SEPVT [2]PPPSD
tidecorr      =      1,0,0,0                           % earth tide correction in PPPSD (0: off, 1: on), [1]solid tides [2]ocean loading [
                                                        % solid tides option must be active to perform any of the other corrections
                                                        % tide corrections requires ERP and ocean loading parameters if activated

satellitepco  =      1,1                               % satellite antenna phase center offset correction (0: off, 1: on) [1]SEPVT [2]PPPS
satellitepcv  =      1                                 % satellite antenna phase center variation correction (0: off, 1: on) [1]PPPSD
receiverpco   =      1,1                               % receiver antenna phase center offset correction (0: off, 1: on) [1]SEPVT [2]PPPSD
receiverpcv   =      1                                 % receiver antenna phase center variation correction (0: off, 1: on) [1]PPPSD
phasewindup   =      1                                 % carrier phase wind up correction (0:off, 1:on) [1]PPPSD
shapirodelay  =      1                                 % gravitational delay correction (0: off, 1: on) for the effect of general relativi
antdelsrc     =      1                                 % the source of receiver antenna delta (0:from obs 1:from opt)
antdel        =      0,0,0                             % receiver antenna delta (m) in ENU frame

csthresmw     =      2                                 % Melbourne-Wubbena sigma factor for MW cycle slip detection threshold
csthresgf     =      0.1,0.5,60                        % parameters for geometry free cycle slip detection threshold ([1]GF maximum thresh
csthresir     =      0.03,0.5,60                       % parameters for ionosphere rate cycle slip detection threshold ([1]IR maximum thre
csthresdi     =      0.85,0.15                         % Doppler integration threshold (cyc)
cs_l_gf       =      8                                 % queue length (number of samples) for geometry-free combination polynomial fit in
csmaxnosn     =      0                                 % maximum allowed samples of signal interruption before resetting CS past data
csminsn       =      1                                 % minimum number of accumulated samples before performing CS detection
maxcscyc      =      0                                 % maximum number of cycles for repairing detected CS
```

Figure 3: Example of GNSS Analysis options in a HAS-UA configuration file.

3.2.3 Filtering options

The complete list of Filtering options with their descriptions and comments is provided in Table 3. An example of the Filtering option part in a configuration file is shown in Figure 4.

Parameter	Description	Values / Notes
wlsmaxiter, wlsconvthresh	Maximum number of iterations and convergence threshold for Weighted Least Squares (WLS) estimation.	
ekfmaxiter, ekfconvthresh	Maximum number of iterations and convergence threshold for the Extended Kalman Filter (EKF).	
ekfppppred	Indicate solution for recovering from intervals with insufficient observations.	0: from SPP. 1: from EKF prediction.
adaptse	Activate the adaptation in DIA for SEPVT (1) and PPPSD (2).	0: off. 1: on.

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Parameter	Description	Values / Notes
lalpha, walpha	Probability of false alarm threshold for respectively for LOM-test and w-test in SEPVT (1) and PPPSD (2).	
avoidWtest	Criteria for computing w-test in SEPVT (1) and PPPSD (2).	0: no w-test performed if LOM passes. 1: w-test performed even if LOM passes.
scalecorrCV	Scale factor for multiplying the variance of the GNSS corrections	
scaleIN_P0	Scale factor for multiplying the variance of the PPPSD parameters on reinitialization.	
scaleIA_P0	Scale factor for multiplying the variance of the newly initialized ionosphere and ambiguity parameters in PPPSD.	
pcnmode	Process noise mode.	0: fixed, provided in pcnstd0. 1: velocity adaptive, provided in pcnstd1a/b/c
pcnstd0	Process noise standard deviation for state Mode 0. (1) pos; (2) vel; (3) acc; (4) clb; (5) cld; (6) tro; (7) ion; (8) amb	Position (pos): m/√s); velocity (vel): m/s/√s; acceleration (acc): $m/s^2/\sqrt{s}$; clock bias (clb): m/√s; clock drift (cld) m/s/√s; troposphere (tro) m/√s; ionosphere (ion) m/√s; ambiguity (amb): cyc/√s clb and cld are not used for PPPSD.
pcnstd1a, pcnstd1b, pcnstd1c	Process noise standard deviations for different ranges (a, b, c) in state Mode 1. For a and b: (1) rx vel; (2) pos; (3) vel; (4) acc; (5) clb; (6) cld; (7) tro; (8) ion; (9) amb. pcnstd1c follows pcnstd0.	Same note as for pcnstd0. Refer to [2] for details.
psrstd	Measurement noise standard deviation for pseudorange (PSR) observations.	
phastd	Measurement noise standard deviation for carrier phase (PHA) observations.	
dplstd	Measurement noise standard deviation for Doppler shift (DPL) measurements.	
meamap	Mapping function coefficients for measurement noise.	
pssstd	Satellite position accuracy noise Normal STD (m). (1) normal, (2) transverse, (3) cross-track.	NTW frame defined in [1].
vlsstd	Satellite velocity accuracy noise Normal STD (m). (1) normal, (2) transverse, (3) cross-track.	NTW frame defined in [1].
cesstd	Satellite clock error accuracy noise Normal std (m). (1) S1, (2) S2.	S1: system 1 (Galileo). S2: system 2 (GPS).
cdsstd	Satellite clock drift accuracy noise Normal std (m/s). (1) S1, (2) S2.	Same note as for cesstd.

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Parameter	Description	Values / Notes
cbsstd	Satellite code bias accuracy noise Normal std (m). (1) S1F1, (2) S1F2, (3) S1F3, (4) S2F1, (5) S2F2, (6) S2F3.	SmFn: signal n of system m as for selected signals in signalList.
pbsstd	Satellite code bias accuracy noise Normal std (m). (1) S1F1, (2) S1F2, (3) S1F3, (4) S2F1, (5) S2F2, (6) S2F3.	SmFn: signal n of system m as for selected signals in signalList.
htrstd	Zenith hydrostatic tropospheric model accuracy noise Normal std (m) for SEPVT (1) and PPPSD (2).	< 0 to use default by model.
wtrstd	Zenith wet tropospheric model accuracy noise Normal std (m) for SEPVT (1) and PPPSD (2).	< 0 to use default by model.
spcostd	Satellite phase center offset accuracy noise Normal std (m)	< 0 to use default by model.
spcvstd	Satellite phase center variation accuracy noise Normal std (m).	< 0 to use default by model.
rpcostd	Receiver phase center offset accuracy noise Normal std (m).	< 0 to use default by model.
rpcvstd	Receiver phase center variation accuracy noise Normal std (m).	< 0 to use default by model.
phwustd	Carrier phase wind up accuracy noise Normal std (m).	< 0 to use default by model.
shapstd	Gravitational delay correction accuracy noise Normal std (m)	< 0 to use default by model.

Table 3: Filtering option list.

```
# Filtering options ***********************************************************************************************************************************
wlsmaxiter     =      25                                  % maximum number of iterations in least squares estimation
wlsconvthresh  =      1e-12                               % residual threshold for WLS convergence

ekfmaxiter     =      0                                   % maximum number of iterations in extended Kalman filter (EKF) estimation
ekfconvthresh  =      1e-3                                % residual threshold for EKF convergence

adaptse        =      1,1                                 % activate (0: off, 1: on) the adaptation in DIA [1]SEPVT [2]PPPSD
lalpha         =      0.05,0.05                           % probability of false alarm threshold for LOM-test [1]SEPVT [2]PPPSD
walpha         =      0.001,0.001                         % probability of false alarm threshold for W-test [1]SEPVT [2]PPPSD
avoidWtest     =      1,1                                 % avoid computing w-test if LOM-test passes (0: no w-test if LOM passes, 1: ye

scaleIA_P0     =      1000                                % scale factor for multiplying the variance of the newly initialized ionospher
pcnmode        =      1                                   % process noise mode (0: fixed -provided in pcnstd0- 1: velocity adaptive -pro
pcnstd0        =      2e-2,2e-2,2e-1,100,1,2e-4,5e-3,0    % mode 0 process-noise square root power spectral density
                                                          % ([1]pos (m/sqrt(s)) [2]vel (m/s/sqrt(s)) [3]acc (m/s^2/sqrt(s))
                                                          %  [4]clb (m/sqrt(s)) [5]cld (m/s/sqrt(s)) [6]tro (m/sqrt(s))
                                                          %  [7]ion (m/sqrt(s)) [8]amb (cyc/sqrt(s))) clb and cld are not used for pppsd
pcnstd1a       =      0.1,2e-2,2e-2,2e-1,100,1,2e-5,5e-3,0 % mode 1 process-noise square root power spectral density 1st range, 0-rx vel
                                                          % ([1]rx vel [2]pos [3]vel [4]acc [5]clb [6]cld [7]tro [8]ion [9]amb) cld is n
pcnstd1b       =      100,1e-2,1e-2,3e0,100,1,2e-4,5e-3,0 % mode 1 process-noise square root power spectral density 2nd range, rx vel la
                                                          % ([1]rx vel [2]pos [3]vel [4]acc [5]clb [6]cld [7]tro [8]ion [9]amb) cld is n
pcnstd1c       =      2e-2,1e-1,3e0,100,1,2e-2,5e-3,0     % mode 1 process-noise square root power spectral density 2nd range, rx vel 1b
                                                          % ([1]pos (m/sqrt(s)) [2]vel [3]acc [4]clb [5]cld [6]tro [7]ion [8]amb) cld is

psrstd         =      0.2,0.1,0,0.2,0.1,0                 % pseudorange measurement noise Normal std (m) [1]S1F1 [2]S1F2 [3]S1F3 [4]S2F1
phastd         =      0.002,0.002,0,0.002,0.002,0         % carrier phase measurement noise Normal std (m) [1]S1F1 [2]S1F2 [3]S1F3 [4]S2
dplstd         =      0.02,0.02,0,0.02,0.02,0             % Doppler measurement noise Normal std (m/s) [1]S1F1 [2]S1F2 [3]S1F3 [4]S2F1 [
meamap         =      1.02,0.02,1.02,0.02                 % measurement mapping function coefficients s(a/sin(ele)+c) ([1]GAL_a [2]GAL_c

pssstd         =      0.001,0.001,0.001                   % satellite position accuracy noise Normal std (m) [1]N [2]T [3]W
vlsstd         =      0.0001,0.0001,0.0001                % satellite velocity accuracy noise Normal std (m/s) [1]N [2]T [3]W
cesstd         =      0.002,0.002                         % satellite clock error accuracy noise Normal std (m) [1]S1 [2]S2
cdsstd         =      0.002,0.002                         % satellite clock drift accuracy noise Normal std (m/s) [1]S1 [2]S2
cbsstd         =      0.002,0.002,0,0.002,0.002,0         % satellite code bias accuracy noise Normal std (m) [1]S1F1 [2]S1F2 [3]S1F3 [4]
pbsstd         =      0.002,0.002,0,0.002,0.002,0         % satellite phase bias accuracy noise Normal std (m) [1]S1F1 [2]S1F2 [3]S1F3 [4
htrstd         =      -1,-1                               % zenith hydrostatic tropospheric model accuracy noise Normal std (m)
                                                          % [1]SEPVT (<0 to use default by model) [2]PPPSD (<0 to use default by model)
wtrstd         =      -1,-1                               % zenith wet tropospheric model accuracy noise Normal std (m)
                                                          % [1]SEPVT (<0 to use default by model) [2]PPPSD (<0 to use default by model)
spcostd        =      -1                                  % satellite phase center offset accuracy noise Normal std (m) (<0 to use defaul
spcvstd        =      -1                                  % satellite phase center variation accuracy noise Normal std (m) (<0 to use def
rpcostd        =      -1                                  % receiver phase center offset accuracy noise Normal std (m) (<0 to use default
rpcvstd        =      -1                                  % receiver phase center variation accuracy noise Normal std (m) (<0 to use defa
phwustd        =      -1                                  % carrier phase wind up accuracy noise Normal std (m) (<0 to use default by mod
shapstd        =      -1                                  % gravitational delay correction accuracy noise Normal std (m) (<0 to use defau
```

Figure 4: Example of Filtering options in a HAS-UA configuration file

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3.2.4 Output options

The complete list of Output options with their descriptions and comments is provided in Table 4. An example of the Output option part in a configuration file is shown in Figure 5.

Parameter	Description	Values / Notes
timef	Output time format.	1: GPS Week + Seconds of Week (SOW). 2: Calendar (YYYY/MM/DD hh:mm:ss).
posf	Output position format.	1: Earth-Centered, Earth-Fixed (ECEF-XYZ) coordinates. 2: Latitude, Longitude, Height (LLH) coordinates.
visualisation	Display the positioning solution during processing.	0: off. 1: on.
logLevel	Sets the log verbosity level for both file and terminal output.	1 (error): only error messages (processing stops). 2 (warning): up to warning messages (potential problems/important notices). 3 (info): up to informational messages (normal system operation). 4 (debug): up to detailed messages useful for debugging.
echoTerminal	Specifies whether log messages are also displayed in the terminal.	0: Disabled. 1: Enabled.

Table 4: Output option list.

```
# output options ***************************************************************************************************************************************
timef          =      1                                  % output time format (1: GPS_week+SOW, 2: Calendar)
posf           =      1                                  % output position format (1: ECEF-XYZ, 2: LLH -latitude longitude height-)
visualisation  =      0                                  % show positioning solution on figure while processing (0: off, 1:on)
logLevel       =      3                                  % Set the log verbosity level for both file and terminal (1: ERROR, 2: WARNING, 3:
echoTerminal   =      1                                  % Enable showing log message in terminal (0: disabled, 1: enabled)
```

Figure 5: Example of Output options in a HAS-UA configuration file.

3.3 HAS-UA execution

The entry point of the HAS-UA software is the *HASUA_Start* function, which serves as the primary interface for users to initiate the processing of their data.

All subdirectories containing the needed functions are added to the MATLAB path before executing the *HASUA_Start* function (e.g. executing the command “addpath(genpath(pwd));” while the working folder is “src”), ensuring that any necessary functions or scripts in these directories can be accessed during execution.

3.3.1 HAS-UA execution example

To execute the HAS-UA from the MATLAB command line, users can utilize a syntax like the one illustrated in Figure 6. This syntax allows users to specify the input configuration files (described in Section 3.2) required for the processing of the data. By running the *HASUA_Start* function with the appropriate input argument, users can launch the HAS-UA and obtain PPP and or SPP solutions. An example of MATLAB script to execute the HAS-UA program is

% clc

tic

clear

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```
% close all
addpath(genpath(pwd));
cngf_flpath = fullfile(pwd, '..', 'conf', 'PPPSD', 'HASUA_PPPSD_MET3.cnf');
HASUA_Start(cngf_flpath);
toc
```

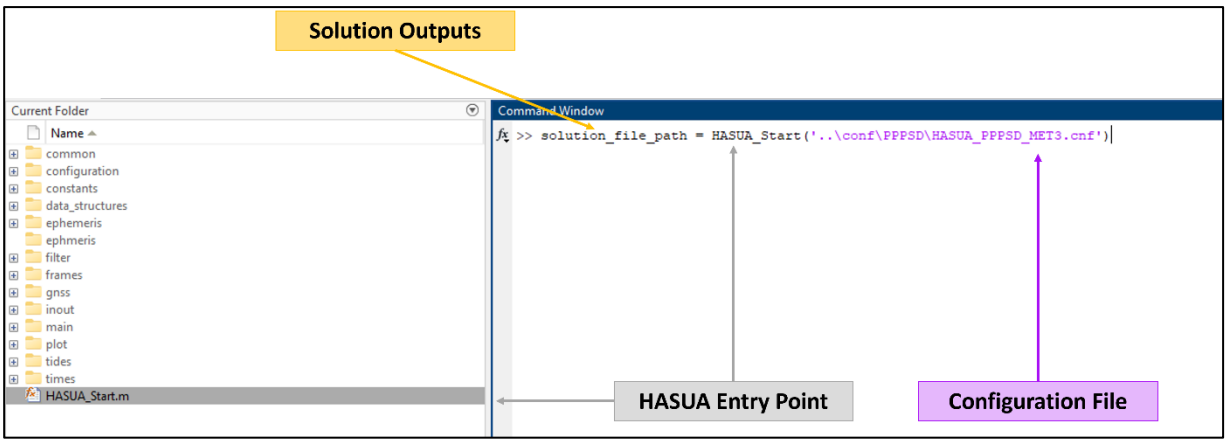


Figure 6: HAS-UA execution example.

3.3.2 HASUA_Start content

The *HASUA_Start* function has three main steps, as shown in Figure 7. The main functionality of the three steps is described below.



Figure 7: *HASUA_Start* steps.

Step 1. Initialize constants, data structures, and default options. The function '*set_default_options*' initializes default environment settings and options, storing them in the "default_environment" and "default_options" structures.

The *set_default_options* function contains a wide range of setting parameters that control various aspects of the HAS-UA processing. The option parameters are described in Section 3.2.

Step 2. Reading the input files. This is the step in which the data are parsed and imported in MATLAB environment using the function *read_input_files*. The *read_input_files* function is designed to read and process various input files related to satellite navigation and positioning, including observation files,

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ephemeris files, clock files, correction files and others. The function returns three output arguments: obsr (a structure containing observation data), nav (a structure containing navigation data – like GNSS satellite ephemerides and clock errors –, correction data and others), and options (a structure with the user defined input options, which are validated with the information from the read input data). The main steps of the reading can follow two ways.

If a mat file containing the processed data already exists and the options allow to use it:

- Load of input data in mat file: load the output structures obsr and nav from the mat file.

Otherwise, the following steps are executed.

- Read broadcast ephemeris file: use the *read_rinex_navigation* function to populate the brd structure.
- Read correction file: use the *read_rtcm_ssr_txt* function to populate the ssr_corrections structure.
- Read earth rotation parameters (ERP) file: use the *read_erp* function to populate the erp structure.
- Read antenna file: use the *read_antex* function to read the antenna file(s) (if two files are specified, from the first it gets only the satellite antennas and from the second only the receiver antenna) and populate the antex structure.
- Read user observation file: use the *read_rinex_observation* function to populate the obsr structure. The *read_rinex_observation* function supports the Receiver Independent Exchange Format (RINEX) format 3.xx.
- (Code commented out) Read ocean tide loading parameter (otlp) file: use the *read_blq_otlp* function to populate the otlp structure. Functionality is not enabled in this version.
- Initialize the nav structure: use the *DataStructuresNAV* function to initialize the nav structures. Refer to the specific function source code for detailed information about the data structure.
- Save processed data to mat file: saves the data to a mat file to speed up subsequent processing of the same dataset.

After either loading the mat file or reading the input files, the following final steps are performed by the function.

- Set APCs: use *set_apcs* function to set the antenna phase center corrections (APCs) for the receiver and the satellites.
- Check of option and input compatibility: verify that the options specified by the user are compatible with the provided input data.

Step 3. Execute the positioning. This step is performed by the *process_gnss* function. The *process_gnss* function is responsible for executing the positioning, utilizing all provided input options, observation data (obsr), navigation, correction, and auxiliary data (nav). The *process_gnss* function achieves this by calling the *solve_gnss* function, the core component of the HAS-UA software, specifically designed to process the GNSS data and generate the navigation solution. The *process_gnss* function accepts various input arguments, including file paths, opt (options structure), obsr, and nav structures. The function returns the file path of the solution file. The main loop over each epoch of the GNSS data is in the *process_gnss* function and the steps performed inside the main loop are outermost interfaces to the core functionality of the HAS-UA software. The functional flow of the *process_gnss* function is the following.

- Initialization: use the *initialize_state_vector* function to initialize the state vector structure and the *calculate_timespan* function to set the processing time span.
- Solution file initialization: use the *output_solution_header* function to initialize the solution file and write its header.
- Main loop: the function enters a main loop, iterating over each epoch of the GNSS data to perform the following set of operations.
    - Search rover observations: use the *search_measurements* function to search for rover observations at the current epoch.
    - Exclude measurements: use the *exclude_satellites* function to discard the measurements according to the configuration options.
    - Search state space representation (SSR) corrections (if SSR correction usage is enabled): use the *search_ssr_corrections* function to search for SSR corrections at the current epoch.

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    - Get ERP: use the *get_erp* function to get the ERP at the current epoch.
    - Sun and Moon position: use the *compute_sun_moon_position* function to calculate the Sun and Moon positions in Earth Centred Earth Fixed (ECEF) coordinates.
    - Satellite positions and clock errors: use the *compute_satellite_states* function to calculate the position, clock bias, velocity, and clock drift of the observed satellites.
    - Solve an epoch: use the *solve_gnss* function to solve for the user navigation solution at the current epoch.
    - Write solution to output file: use the *output_solution* function to write the solution to the output file.
    - Store data in result matrices: the solution data is stored in the result matrices, results_sepvt and results_pppsd.
- Write solution to output mat file: after the main loop, use the *write_solution_output* function to save the solution (variables opt, files, obsr_nsat, ssr_nsat, svs_nsat, results_sepvt and statevec) in an output mat file.
- Close files: finally, use the *close_files* function to close the output and log files.

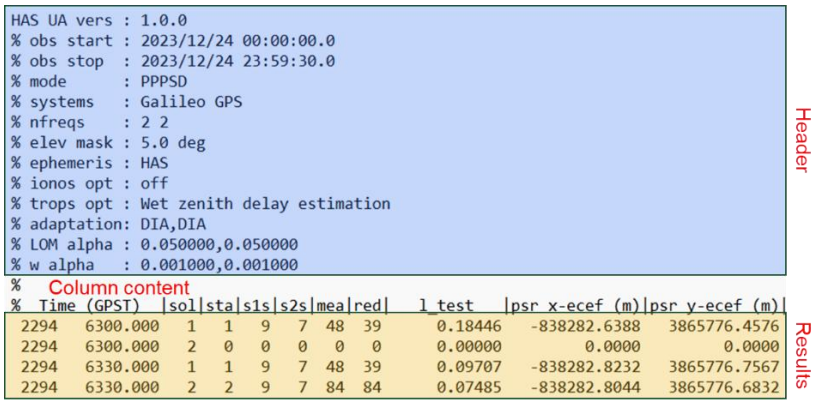
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3.4 Output File

The HAS UA MATLAB version provides three output files:

- a text file including the positioning results (.sol);
- MATLAB workspace at the end of the processing, only for MATLAB version (.mat);
- a text file containing the messages of the software (.log).

The text file has the extension .sol and it is a text file reporting the results generated with the HAS UA. The file is composed of two parts, the header and the results, as shown in Figure 8. In the lower part of Figure 8, the list of the column header is shown.




```
%  Time (GPST)  |sol|sta|s1s|s2s|mea|red|   l_test   |psr x-ecef (m)|psr y-ecef (m)|psr z-ecef (m)|
vlr x(m/s)|vlr y(m/s)|vlr z(m/s)|acr x(m/s^2)|acr y(m/s^2)|acr z(m/s^2)|  cbr S01 (m) |
cbr S02 (m) |   cdr (m)    |   zwt (m)    |psrSx(m)|psrSy(m)|psrSz(m)|vlrSx(m/s)|vlrSy(m/s)|
vlrSz(m/s)|acrSx(m/s^2)|acrSy(m/s^2)|acrSz(m/s^2)|cbrSS01(m)|cbrSS02(m)|cdrS (m)|zwtS (m)|
```

Figure 8: HAS-UA output .sol example.

The header provides metadata that describes the context and options used for the data processing. The description of the header fields is provided in Table 5.

Field	Description	Context / Notes
HAS UA vers	The version of the software or algorithm used for data processing.	
obs start / obs stop	The start and stop times defining the observation period.	
mode	Specifies the mode of operation used to generate the positioning solution.	PPPSD: PPPSD selected. SEPVT: SEPVT selected.
systems	The satellite navigation systems used during processing.	Galileo Only, GPS Only, Galileo and GPS.
nfreqs	The number of signals per system used for the measurements. (1) Galileo, (2) GPS.	
elev mask	The set elevation mask angle (in degrees).	
ephemeris	Indicates the type of ephemeris data used.	This field reflects the user's selection in the configuration file (sateph field).

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Field	Description	Context / Notes
ionos opt / trops opt	Options denoting how ionospheric and tropospheric delays were handled (models/corrections used).	This field reflects the user's selection in the configuration file (trowest field). Ionos opt always off.
adaptation	Methods of adaptation used during processing for SEPVT (1) and PPPSD (2), respectively (e.g., for statistical testing).	DI: no adaptation performed. DIA: adaptation performed
LOM alpha / w alpha	Probability of false alarm related to the DIA (Detection Identification and Adaptation) algorithm for SEPVT (1) and PPPSD (2), respectively.	This field reflects the user's selection in the configuration file (lalpha, walpha field).

Table 5: HAS-UA solution file header fields.

The results for each epoch are reported in the .sol file and they are organized in columns. Each row in the data section represents a solution of an active estimation method. The description of the HAS UA solution file (.sol) data columns is provided in Table 6.

Column Name	Description	Values / Units
Time & Solution Status		
Time (GPST)	Measurement time in GPS Time (GPST).	GPS Week + Time of Week (ToW) in seconds (starting midnight Sunday).
sol	Positioning Solution Mode used for estimation.	1: SEPVT mode. 2: PPPSD mode.
sta	Solution Status Indicator.	0: Solution not available. 1: Solution available in SPP mode. 2: Solution available in PPP mode.
Statistics & Measurements		
s1s, s2s	Number of satellites used in the solution for Galileo (s1s) and GPS (s2s), respectively.	Integer Count
mea, red	Number of measurements and redundancy, respectively.	Integer Count
l_test	LOM test value from DIA	Unitless
Estimated State Vectors (Position, Velocity, Acceleration)		
psr x-ecef, y-ecef, z-ecef	Position Estimation in Earth-Centered Earth-Fixed (ECEF) coordinates.	(m)
vlr x, vlr y, vlr z	Velocity components (receiver's velocity) in ECEF coordinates.	(m/s)
acr x, acr y, acr z	Acceleration components in ECEF coordinates.	(m/s^2)
Estimated State Vectors (Clocks & Atmosphere)		
cbr S01, cbr S02	Receiver Clock Bias with respect to different GNSS time scales (S01 Galileo, S02 GPS).	(m)
cdr	Receiver Clock Drift (rate of change of the clock bias).	(m/s)
zwt	Estimated Zenith Wet Tropospheric Delay (ZWD).	(m)
Estimated State Variances (Position, Velocity, Acceleration)		

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Column Name	Description	Values / Units
psrSx, psrSy, psrSz	Formal standard deviation of the estimated ECEF Position coordinates.	(m)
vlrSx, vlrSy, vlrSz	Formal standard deviation of the estimated ECEF Velocity components.	(m/s)
acrSx, acrSy, acrSz	Formal standard deviation of the estimated ECEF Acceleration components.	(m/s^2)
Estimated State Variance (Clocks & Atmosphere)		
cbrSS01, cbrSS02	Formal standard deviation of the estimated receiver Clock Biases.	(m)
cdrS	Formal standard deviation of the estimated receiver Clock Drift.	(m/s)
zwtS	Formal standard deviation of the estimated ZWTD (Zenith Wet Tropospheric Delay).	(m)

Table 6: Description of the HAS UA solution file (.sol) data columns.

The MATLAB workspace at the end of the processing is saved in a .mat file, which is provided to the user for a more in-depth evaluation, an example of the workspace is shown in Figure 9.

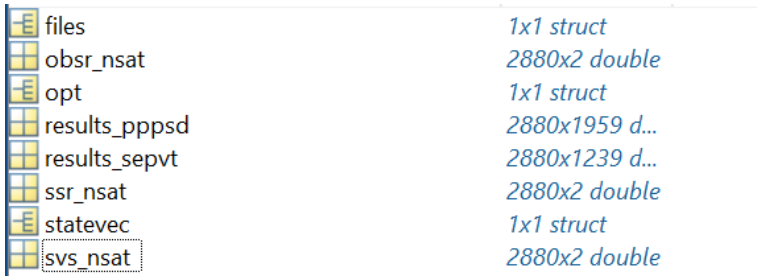


Figure 9: HAS-UA output example .mat case.

The .mat file includes the following elements:

- files (structure): contains information about the files used in the processing, including file names and paths.
- opt (structure): contains the options selected for the processing.
- statevec (structure): holds information related to the state vector.
- obsr_nsat (matrix): a numerical matrix with dimensions corresponding to the number of epochs by two. It indicates the number of satellites from the observation file, with the first column showing the number of Galileo satellites and the second column showing the number of GPS satellites.
- ssr_nsat (matrix): a numerical matrix with dimensions corresponding to the number of epochs by two. It indicates the number of satellites from the SSR correction file, with the first column showing the number of Galileo satellites and the second column showing the number of GPS satellites.
- svs_nsat (matrix): a numerical matrix with dimensions corresponding to the number of epochs by two. It indicates the number of satellites used in the navigation solution, with the first column showing the number of Galileo satellites and the second column showing the number of GPS satellites.
- results_pppsd (matrix): Reports the results of the PPPSD engine, with each row corresponding to a specific processing epoch. The results_pppsd matrix columns are described in Table 7.
- results_sepvt (matrix): Reports the results of the SEPVT engine, with each row corresponding to a specific processing epoch. The results_sepvt matrix columns are described in Table 7.

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Column Field	Description	Unit / Notes
Time & Metadata		
GPS week number	The current GPS Week number.	Integer
Time of Week	Time elapsed since the start of the week (Sunday midnight GPST).	(s)
Solution status flag	A bit field (0-7) indicating the status of the solution at the epoch.	Bit 0: solution is based on SEPVT. Bit 1: solution is based on PPPSD. Bit 2: reserved. Bit 3: EKF predicted solution. Bit 4: detection, LOM test over threshold. Bit 5: identification w-test over threshold. Bit 6: adaptation, solution adapted. Bit 7: imaginary w-test.
Pivot Sat ID (Gal)	Satellite ID of the pivot satellite for Galileo.	Satellite ID (01-40)
Pivot Sat ID (GPS)	Satellite ID of the pivot satellite for GPS.	Satellite ID (41-80)
Used Satellites (Gal)	Number of used Galileo satellites.	Integer Count
Used Satellites (GPS)	Number of used GPS satellites.	Integer Count
Number of measurements	Total number of measurements available at the epoch.	Integer Count
Redundancy value	Redundancy value of the system, as defined in the referenced source [2].	Unitless
Value of the LOM test	Value of the LOM test (from DIA).	Unitless
Estimated State Vector Parameters (Values)		
Estimated coordinates (3 columns)	Estimated receiver position in ECEF coordinates (r_x, r_y, r_z).	(m) No solid tide displacement applied.
Estimated velocity (3 columns)	Estimated receiver velocity in ECEF components (v_x, v_y, v_z).	(m/s)
Estimated acceleration (3 columns)	Estimated receiver acceleration in ECEF components (a_x, a_y, a_z).	(m/s^2)
Estimated clock components (3 columns)	Estimated receiver clock bias per system and drift.	(m) or (m/s) All zeros in PPPSD as parameters not estimated.
Estimated zenith wet tropospheric delay (ZWD)	Estimated Zenith Wet Tropospheric Delay.	(m) All zeros in SEPVT as parameters not estimated.

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Column Field	Description	Unit / Notes
Estimated slant ionospheric delay	Estimated slant ionospheric delay parameters.	Number of columns depends on used constellations (40 satellites per active constellation are allocated). All zeros in SEPVT as parameters not estimated.
Estimated float ambiguities	Estimated float ambiguity values.	Number of columns depends on used constellations times the activated frequencies (40 satellites per active constellation per frequency). All zeros in SEPVT as parameters not estimated.
Estimated State Vector Parameters (Standard Deviations)		
Variances of the state vectors parameters	The standard deviation for all estimated state parameters reported in the same sequential order.	Same units of state parameters.
Residuals and Testing		
w-test values	Statistical w-test values.	Same number of columns as float ambiguities for each measurement type activated in SEPVT or PPPSD. Non-zero only if detection LOM test is above threshold or w-test forced by option.
post-fitting residuals ($\hat{e}_{r,j}^{s}$)	The post-fit measurement residuals, as defined in [2].	Same number of columns as the w-test.
predicted residuals ($v_{r,j}^{s}$)	The predicted measurement residuals, as defined in [2].	Same number of columns as the w-test. All zeros in SEPVT as not computed.
Solid Tide Corrections		
solid tide displacement (x, y, z GTRF)	The displacement vector applied for the solid Earth tides, reported in the GTRF (Galileo Terrestrial Reference Frame).	(m) Computed only if PPPSD mode is selected.

Table 7: Description of the results_pppsd and results_sepvt data columns.

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3.5 Sample results

In this section, a sample result obtained with HAS UA algorithm is shown. Specifically, one day of data from the IGS station MET3 is considered. The processing is carried out in PPPSD mode, considering Galileo and GPS in dual frequency configuration (considering the frequency E1 E5a for Galileo and L1 L2 for GPS). All the configuration options for the case HASUA_PPPSD_MET3_2023_358_E1_E5a_L1_L2_SIS.cnf are provided in the attached configuration file.

HASUA_PPPSD_MET
3_2023_358_E1_E5a_

The time evolution of the horizontal and vertical position errors is shown in the upper and lower part of Figure 10, respectively. The East and North errors are shown in the scatter plot in Figure 11; the errors are computed with respect to the reference position of the station retrieved from IGS SINEX file.

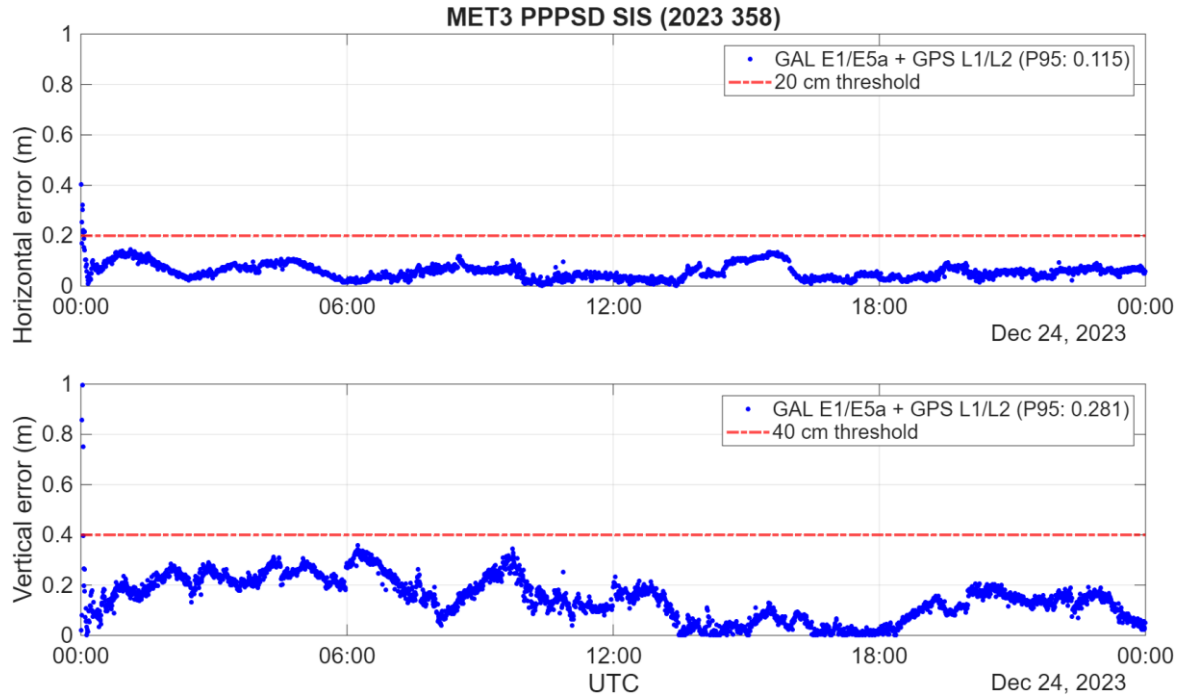


Figure 10: Time series of the HAS-UA results for the MET3 station, year 2023 DOY 358, Galileo and GPS dual frequency (E1-E5a/L1-L2) using SIS corrections. Horizontal Error in the upper part; Vertical Error in the lower part.

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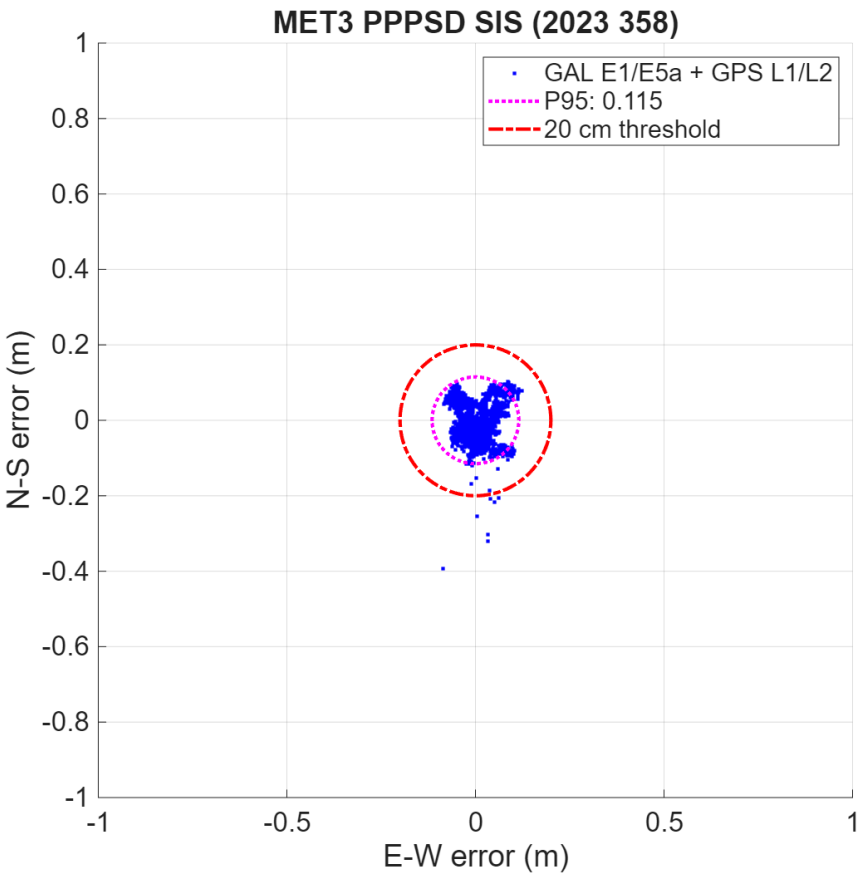


Figure 11: Scatter plot of the HAS-UA results for the MET3 station, year 2023 DOY 358, Galileo and GPS dual frequency (E1-E5a/L1-L2) using SIS corrections. East and North error computed with respect to the reference coordinates from IGS SINEX.

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